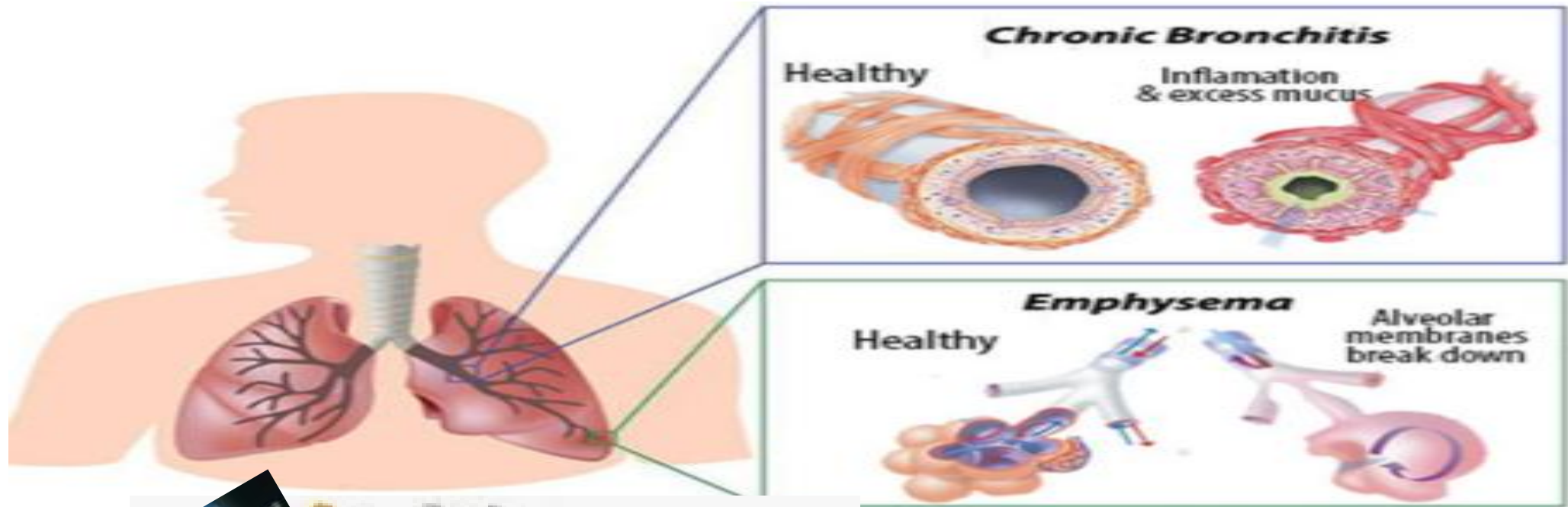


Pharmacotherapy of Chronic Obstructive Pulmonary Disease(COPD)



By;
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Learning objectives

- Upon completion of the chapter, the students will be able to:
 - ❖ Describe the etiology, and pathophysiology of chronic obstructive pulmonary disease (COPD).
 - ❖ Identify signs and symptoms of COPD.
 - ❖ List the treatment goals for a patient with COPD.
 - ❖ Design an appropriate COPD maintenance treatment regimen based on patient-specific data.
 - ❖ Design an appropriate COPD exacerbation treatment regimen based on patient-specific data.
 - ❖ Develop a monitoring plan to assess effectiveness and adverse effects of pharmacotherapy for COPD.
 - ❖ Formulate an appropriate education plan for a patient with COPD.

Introduction

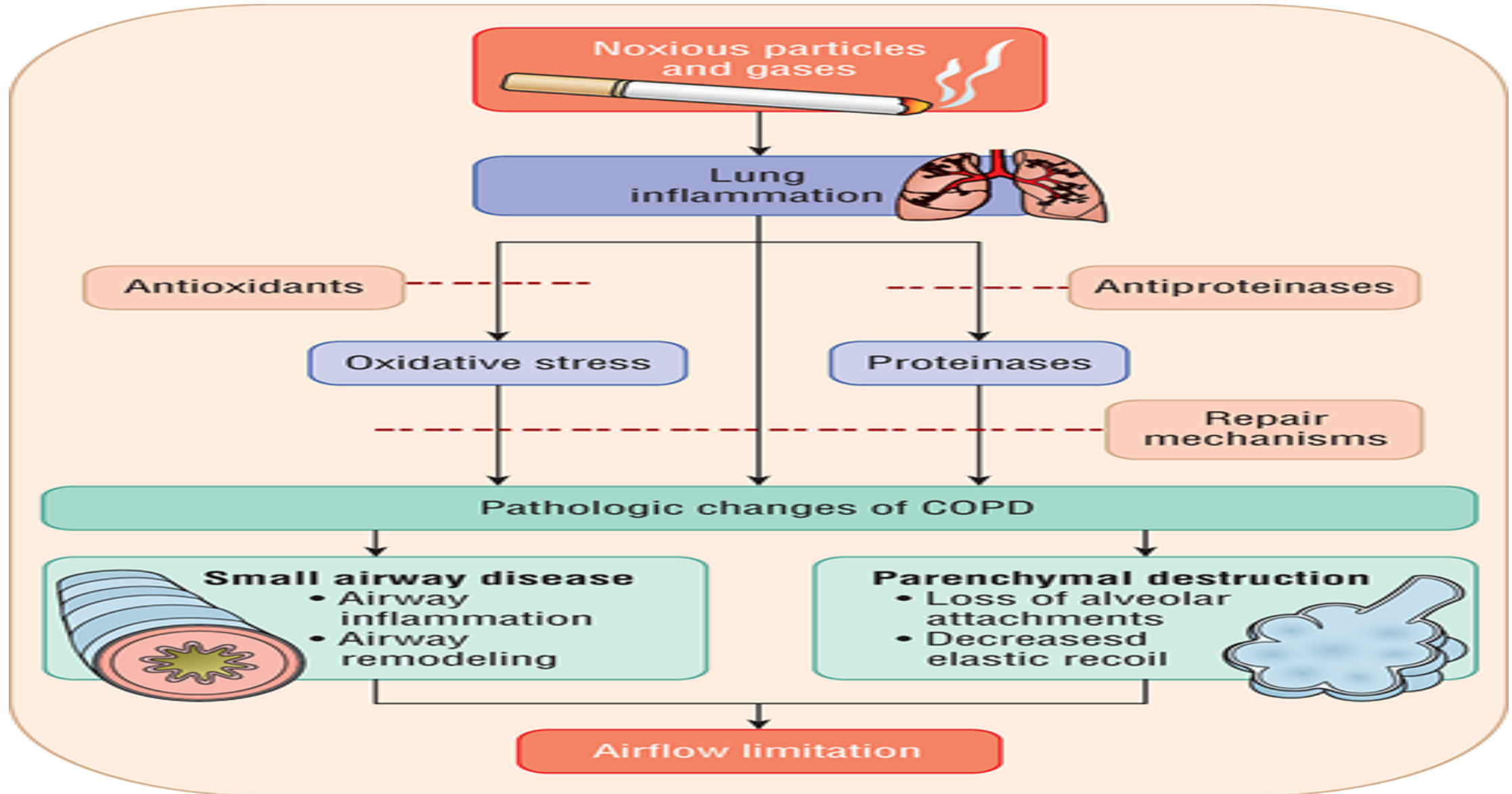
Chronic obstructive pulmonary disease (COPD)

- A common lung disease characterized by airflow limitation that is not fully reversible, in contrast to the reversibility of airflow limitation in asthma.
- It is both **chronic and progressive** and is associated with an abnormal inflammatory response of the lungs to noxious particles or gases.
- ❖ COPD is preventable and treatable and causes significant extrapulmonary effects that contribute to disease severity in a subset of patients.
- ❖ **Exacerbations and comorbidities** contribute to the overall severity in individual patients.

❑ Two principal conditions (referred to as phenotypes) include:

- ❖ ***Chronic bronchitis***: chronic or recurrent excess mucus secretion with cough that occurs on most days for at least 3 months of the year for at least 2 consecutive years.
- ❖ ***Emphysema***: abnormal, permanent enlargement of the airspaces distal to the terminal bronchioles, accompanied by destruction of their walls, without fibrosis.

Mechanisms Underlying Airflow Limitation in COPD



Source: Joseph T. DiPiro, Gary C. Yee, Stuart T. Haines, Thomas D. Nolin, Vicki L. Ellingrod, L. Michael Posey: *DiPiro's Pharmacotherapy: A Pathophysiologic Approach, 12e* Copyright © McGraw Hill. All rights reserved.

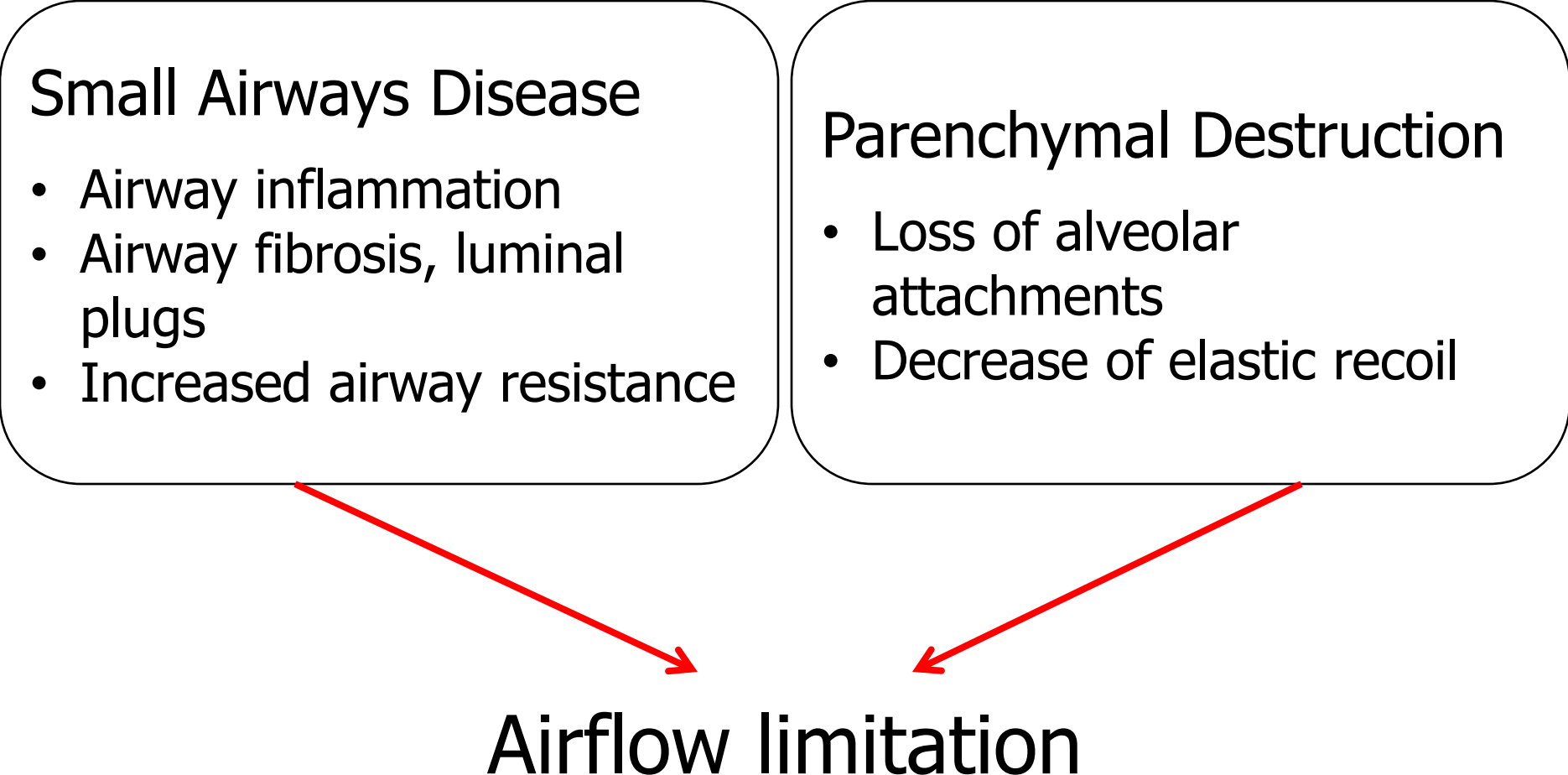
Mechanisms Underlying Airflow Limitation in COPD

Small Airways Disease

- Airway inflammation
- Airway fibrosis, luminal plugs
- Increased airway resistance

Parenchymal Destruction

- Loss of alveolar attachments
- Decrease of elastic recoil



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graph TD; A[Small Airways Disease] --> D[Airflow limitation]; B[Parenchymal Destruction] --> D;
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Airflow limitation

Epidemiology

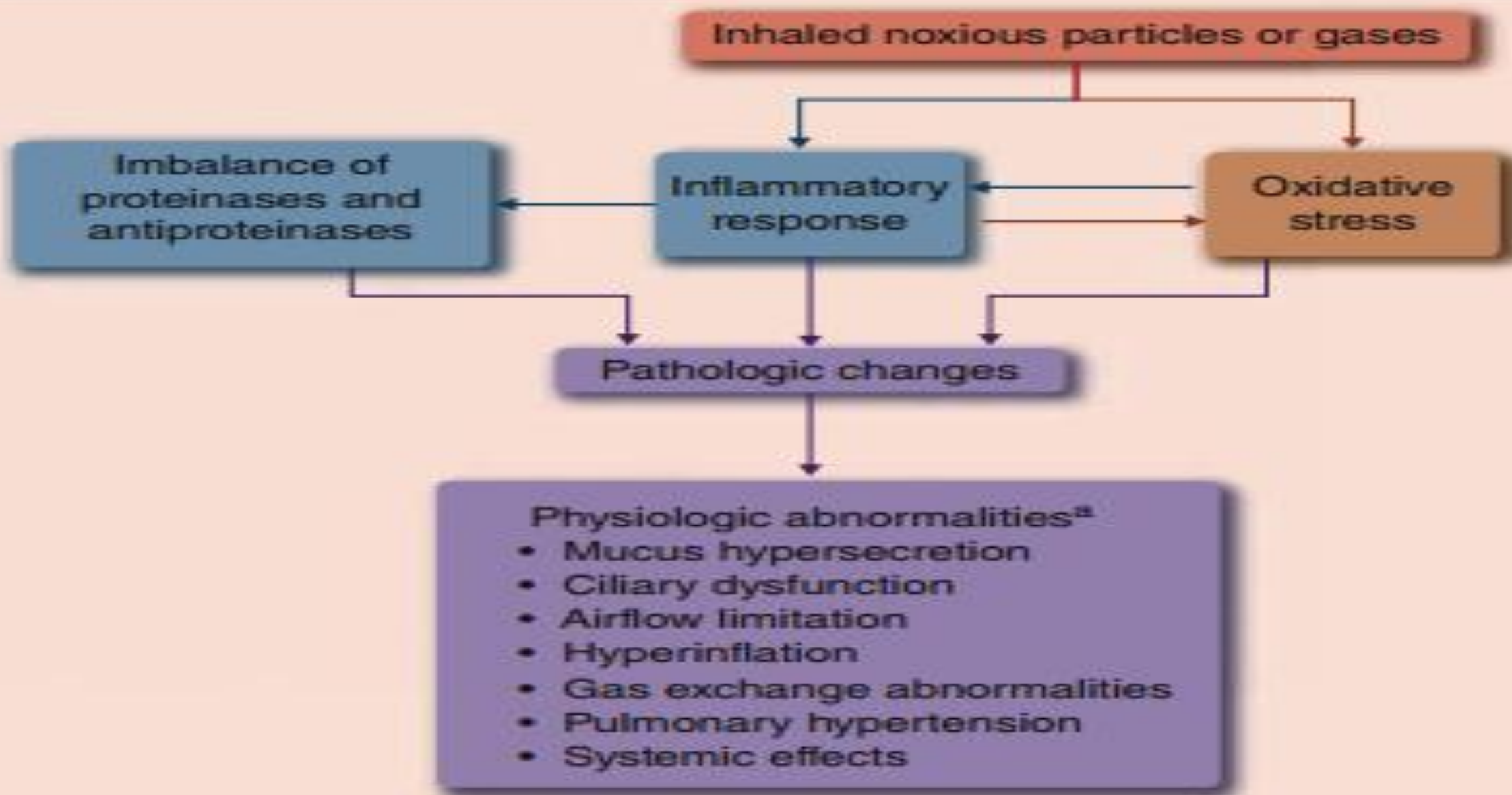
- COPD is a leading cause of morbidity and mortality worldwide.
- The burden of COPD is projected to increase in coming decades due to continued exposure to **COPD risk factors and the aging** of the world's population.
- COPD is associated with significant economic burden.
- COPD is the fourth leading cause of death in the US, exceeded only by cancer, heart disease, and unintentional injuries.
- Over 150,000 deaths are attributed to COPD annually.

Risk Factors for COPD

- ❖ Genes
- ❖ Exposure to particles
 - Tobacco smoke(accounts for 85% to 90% of cases of COPD)
 - Occupational dusts, organic and inorganic
 - Indoor air pollution from heating and cooking with biomass in poorly ventilated dwellings
 - Outdoor air pollution
- ❖ Lung growth and development
- ❖ Age
- ❖ Respiratory infections
- ❖ Socioeconomic status
- ❖ Asthma/Bronchial hyperreactivity
- ❖ Chronic Bronchitis

Pathophysiology

- Repeated exposure to noxious particles and gases causes chronic inflammation, resulting in pathologic changes in the central and peripheral airways, lung parenchyma, and pulmonary vasculature that lead to obstruction.
- An imbalance between **proteinases and antiproteinases** in the lungs and oxidative stress are also important in the pathogenesis of COPD



^aThe physiologic abnormalities usually develop in this order.

Features of Inflammation in COPD Compared with Asthma

	COPD	Asthma
Cells	•Neutrophils •Large increase in macrophages •Increase in CD8+ T lymphocytes	•Eosinophils •Small increase in macrophages •Increase in CD4+ Th2 lymphocytes •Activation of mast cells
Mediators	•LTB4 •IL-8 •TNF-α	•LTD4 •IL-4, IL-5 •(Plus many others)
Consequences	•Squamous metaplasia of the epithelium •Parenchymal destruction •Mucus metaplasia •Glandular enlargement	•Fragile epithelium •Thickening of basement membrane •Mucus metaplasia •Glandular enlargement
Response to treatment	•Glucocorticosteroids have a variable effect	•Glucocorticosteroids inhibit inflammation

Clinical presentation

- ❖ Initial symptoms include **chronic cough** and **sputum production**; patients may experience cough for several years before **dyspnea** develops.
- ❖ Physical examination is normal in most patients in milder stages.
 - When airflow limitation becomes severe, patients may have
 - ✓ cyanosis of mucosal membranes,
 - ✓ development of a “barrel chest” due to hyperinflation of the lungs,
 - ✓ increased resting respiratory rate,
 - ✓ shallow breathing,
 - ✓ pursing of lips during expiration, and
 - ✓ use of accessory respiratory muscles.

Exacerbation of COPD

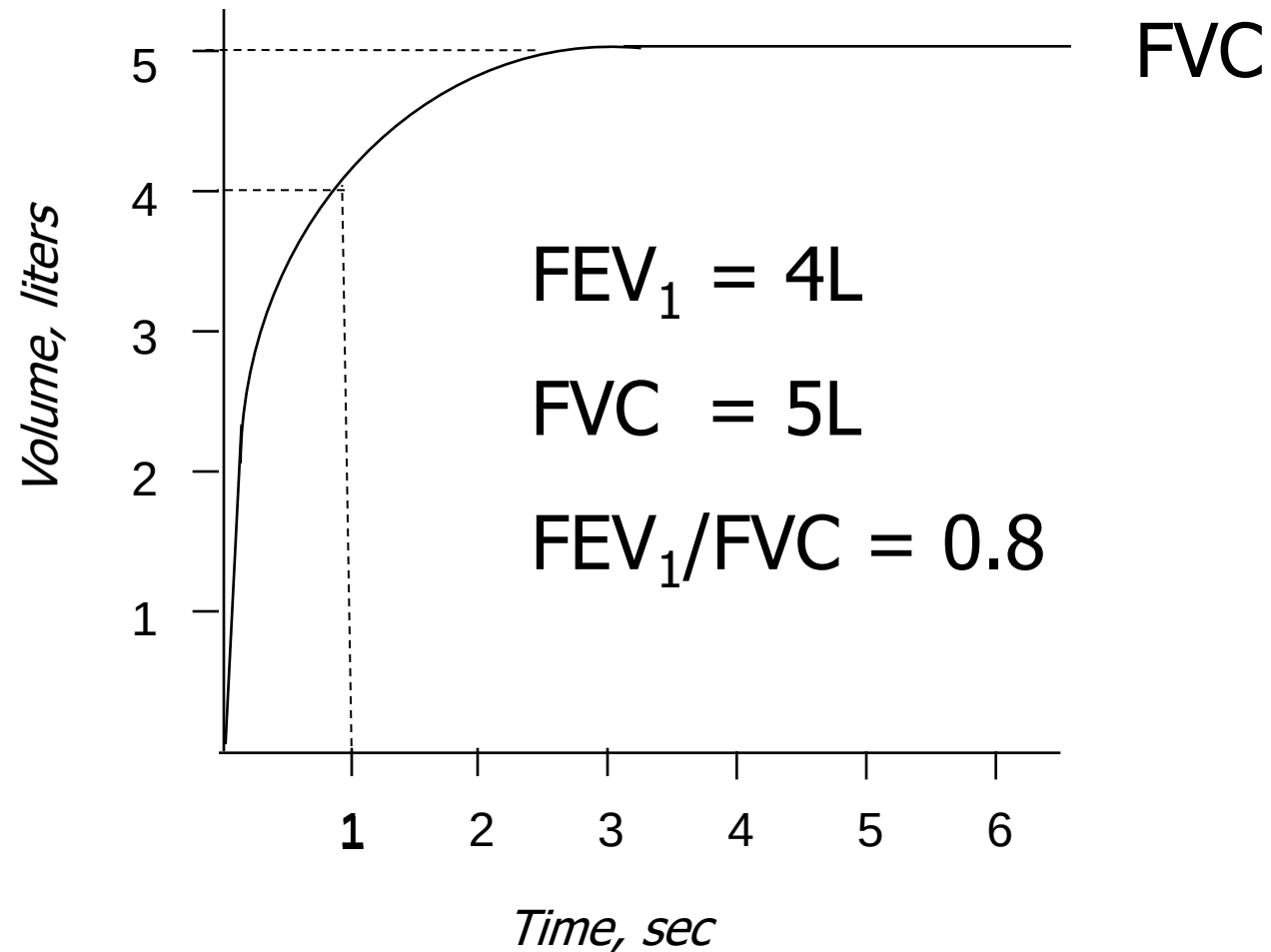
- It is an acute worsening of respiratory symptoms that results in additional therapy.
- The cause of acute exacerbations could be a result of **respiratory tract infections**, either **viral** or **bacterial**, **air pollution**, or other environmental exposures.
- These triggers can initiate increased bronchospasm and airway resistance in an individual with already compromised lung function.

- Patients experiencing COPD exacerbation may have **worsening dyspnea, increased sputum volume, or increased sputum purulence.**
 - Other features of exacerbation include
 - ✓ chest tightness,
 - ✓ increased need for bronchodilators,
 - ✓ malaise,
 - ✓ fatigue, and
 - ✓ decreased exercise tolerance.

Diagnosis

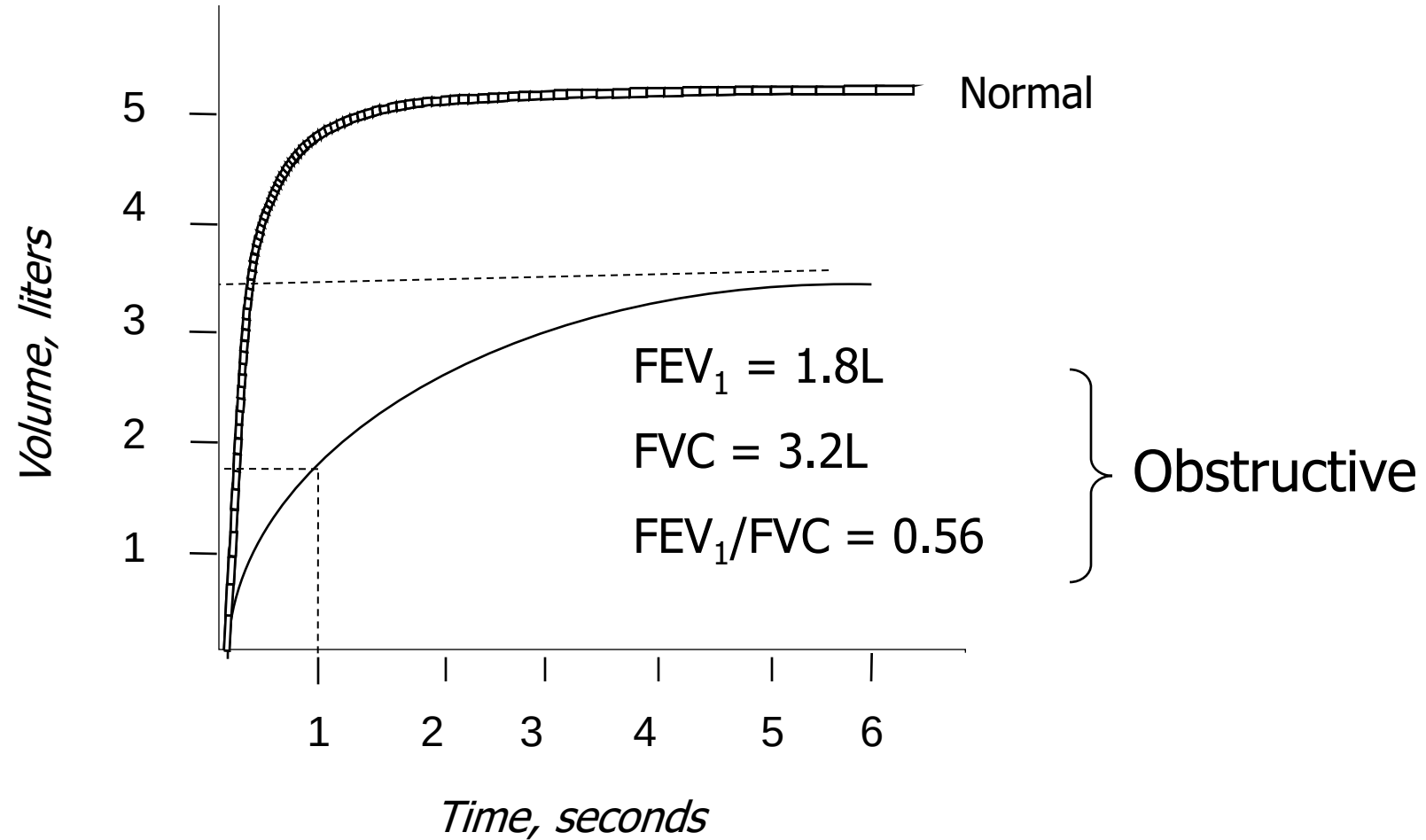
- Symptoms(dyspnea, chronic cough or sputum production) and/or
- History of exposure to risk factors
- Spirometry is *required* to make the diagnosis;
 - The presence of a post-bronchodilator $FEV_1/FVC < 0.70$ confirms the presence of persistent airflow limitation and thus of COPD.
- Severity classification is also required

Spirometry: Normal Trace Showing FEV₁ and FVC



FEV1: Forced expiratory volume in one second ; FVC: forced vital capacity

Spirometry: Obstructive Disease



Classification of Severity of Airflow Limitation in COPD*

In patients with $FEV_1/FVC < 0.70$:

GOLD 1: Mild $FEV_1 \geq 80\%$ predicted

GOLD 2: Moderate $50\% \leq FEV_1 < 80\%$ predicted

GOLD 3: Severe $30\% \leq FEV_1 < 50\%$ predicted

GOLD 4: Very Severe $FEV_1 < 30\%$ predicted

**Based on Post-Bronchodilator FEV_1*

Assess Risk of Exacerbations

To assess risk of exacerbations use history of exacerbations and spirometry:

- Two or more exacerbations within the last year *or*
- $FEV_1 < 50\%$ of predicted value are indicators of high risk.

Assess COPD Comorbidities

COPD patients are at increased risk for:

- Cardiovascular diseases
- Osteoporosis
- Respiratory infections
- Anxiety and Depression
- Diabetes
- Lung cancer

❖ *These comorbid conditions may influence mortality and hospitalizations and should be looked for routinely, and treated appropriately.*

Differential Diagnosis: COPD and Asthma

COPD

- Onset in mid-life
- Symptoms slowly progressive
- Long smoking history

ASTHMA

- Onset early in life (often childhood)
- Symptoms vary from day to day
- Symptoms worse at night/early morning
- Allergy, rhinitis, and/or eczema also present
- Family history of asthma

Treatment

Desired Outcomes

- The goals of COPD management include:
 - ❖ Smoking cessation if applicable,
 - ❖ Reducing symptoms,
 - ❖ Improving exercise tolerance,
 - ❖ Minimizing the rate of decline in lung function,
 - ❖ Maintaining or improving quality of life, and
 - ❖ Preventing and treating exacerbations.

Non-Pharmacologic Therapy

❖ Patients should receive education about their disease, treatment plans, and strategies to slow progression and prevent complications.

☐ **Smoking Cessation**

- Smoking cessation is the most important intervention to prevent development and progression of COPD.
- Reducing exposure to occupational dust and fumes as well as other environmental toxins is also important.

First-Line Pharmacotherapies for Smoking Cessation

Agent	Usual Dose	Duration	Common Complaints
Bupropion SR	150 mg orally daily for 3 days, then twice daily	12 weeks, up to 6 months	Insomnia, dry mouth
Nicotine gum or lozenge	2-4 mg prn, up to 24 pieces daily	12 weeks	Sore mouth, dyspepsia
Nicotine inhaler	6-16 cartridges daily	Up to 6 months	Sore mouth and throat
Nicotine nasal spray	8-40 doses daily	3-6 months	Nasal irritation
Nicotine patches	Various, 7-21 mg every 24 hr	Up to 8 weeks	Skin reaction, insomnia
Varenicline	0.5 mg daily orally for 3 days, then 0.5 mg twice daily for 4 days, then 1 mg twice daily	12 weeks	Nausea, sleep disturbances

❑ Pulmonary Rehabilitation

- Pulmonary rehabilitation programs include exercise training, breathing exercises, optimal medical treatment, psychosocial support, and health education.

❑ Immunizations

- ✓ Administer the influenza vaccine annually during influenza season.
- ✓ The CDC recommends giving the 23-valent pneumococcal polysaccharide vaccine (PPSV23) for people from ages 2 to 64 who have chronic lung disease, smokers over the age of 18, and all people older than 65 years.
- ✓ The GOLD guidelines recommend the pneumococcal vaccine for COPD patients less than age 65 only if the FEV1 is less than 40% predicted

❑ Long-Term Oxygen Therapy

- ❖ Use of supplemental oxygen therapy increases survival in COPD patients with chronic hypoxemia at rest.
- ❖ Once patients are stabilized as outpatients and pharmacotherapy is optimized, institute long-term oxygen therapy if either
 1. resting PaO₂ less than 55 mm Hg or SaO₂ less than 88% with or without hypercapnia, or
 2. resting PaO₂ 55 to 60 mm Hg or SaO₂ less than 88% with evidence of right-sided heart failure, polycythemia, or pulmonary hypertension.
 - The goal is to raise PaO₂ above 60 mm Hg.

□Surgery

- Bullectomy, lung volume reduction surgery, and lung transplantation are surgical options for very severe COPD.
- These procedures may result in improved spirometry, lung volumes, exercise capacity, dyspnea, health-related quality of life, and possibly survival.
- Patient selection is critical because not all patients benefit.

Pharmacologic Therapy of COPD

Beta ₂ -agonists
Short-acting beta ₂ -agonists
Long-acting beta ₂ -agonists
Anticholinergics
Short-acting anticholinergics
Long-acting anticholinergics
Combination short-acting beta ₂ -agonists + anticholinergic in one inhaler
Methylxanthines
Inhaled corticosteroids
Combination long-acting beta ₂ -agonists + corticosteroids in one inhaler
Systemic corticosteroids
Phosphodiesterase-4 inhibitors

	Medication	Onset	Peak	Duration	Usual Dose
Short-Acting β_2 -Agonists	Albuterol^a Nebulization	5–8 min	1–2 hours	3–6 hours	2.5 mg every 4–8 hours (max: 30 mg/day)
	Inhalation	5–8 min	0.5–1 hour	3–6 hours	MDI (90 mcg/puff) one to two puffs every 4–6 hours (max: 1080 mcg/day)
	Oral	7–30 min	2–3 hours	6–8 hours ER: 8–12 hours	2–4 mg three to four times a day ER: 4–8 mg every 12 hours (max: 32 mg/day)
	Levalbuterol Nebulization	10–20 min	1.5 hours	5–8 hours	0.63–1.25 mg three times per day, 6–8 hours apart (max: 3.75 mg/day)
	Inhalation	5–10 min	1–1.5 hours	3–6 hours	MDI (45 mcg/puff) one to two puffs every 4–6 hours (max: 540 mcg/day)
Long-Acting β_2 -Agonists	Terbutaline Oral	0.5–2 hours	1–3 hours	6–8 hours	2.5–5 mg three times per day, 6 hours apart ^b (max: 15 mg/day)
	Formoterol Inhalation	1–3 min	1–3 hours	8–12 hours	Powder (12 mcg/inhalation) one inhalation every 12 hours (max: 24 mcg/day)
	Nebulization	1–3 min	1–3 hours	8–12 hours	20 mcg every 12 hours (max: 40 mcg/day)
	Salmeterol Inhalation	10 min to 1 hour	2–3 hours	12 hours	Powder (50 mcg/inhalation) one inhalation every 12 hours (max: 100 mcg/day)
	Indacaterol Inhalation	5 min	1–4 hours	24 hours	Powder (75 mcg/inhalation) one inhalation every 24 hours (max: 75 mcg/day)
	Olodaterol Inhalation ^c	5 min	—	24 hours	2.5 mcg/inhalation Two inhalations every 24 hours (max: 5 mcg/day)
	Vilanterol Inhalation	15–30 min	1 hour	24 hours	Powder (25 mcg/inhalation) one inhalation every 24 hours (max: 25 mcg/day)
	Arformoterol Nebulization	7–20 min	1–3 hours	12 hours	15 mcg every 12 hours (max: 30 mcg/day)

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Short-Acting Anticholinergic	Ipratropium				
	Nebulization	15 min	1–2 hours	4–8 hours	500 mcg every 6–8 hours (max: 2,000 mcg/day)
	Inhalation	15 min	1–2 hours	4–8 hours	MDI (18 mcg/puff) two puffs four times/day (max: 216 mcg/day)
Long-Acting Anticholinergic	Tiotropium				
	Inhalation	30 min	1–4 hours	24 hours	Powder (18 mcg/inhalation) one inhalation every 24 hours (max: 18 mcg/day) ^d Aerosol solution (2.5 mcg/inhalation) two inhalations every 24 hours (max 5 mcg/day) ^d
	Aclidinium				
	Inhalation	30 min	1–4 hours	12 hours	Powder (400 mcg/inhalation) one inhalation every 12 hours (max: 800 mcg/day)
	Umeclidinium				
	Inhalation	30 min	1–3 hours	24 hours	Powder (62.5 mcg/inhalation) one inhalation every 24 hours (max: 62.5 mcg/day)
Methylxanthine	Theophylline				
	Oral	0.5–2 hours	Up to 24 hours, depending on formulation	6–24 hours	400–600 mg/day divided every 6–24 hours based on formulation (max: 800 mg/day) Adjust dose to serum concentrations of 5–15 mcg/mL (5–15 mg/L; 28–83 µmol/L)
Phosphodiesterase-4 Inhibitor	Roflumilast	4 wks	—	—	500 mcg daily ^e
	Oral				

(Continued)

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	Medication	Onset	Peak	Duration	Usual Dose
Inhaled Corticosteroids	Beclomethasone	1–7 days	1–4 wks		MDI (40, 80 mcg/puff) 40–160 mcg twice a day (max: 640 mcg/day)
	Budesonide	1–7 days	1–2 wks		Powder (90, 180 mcg/inhalation) 180–360 mcg twice a day (max: 1440 mcg/day)
	Ciclesonide	1–7 days	1–4 wks		MDI (80, 160 mcg/puff) 80–320 mcg one or two times per day (max: 640 mcg/day)
	Fluticasone	1–7 days	1–2 wks		MDI (44, 110, 220 mcg/puff) 88–440 mcg twice a day (max: 1760 mcg/day) Powder (50, 100, 250 mcg/inhalation) 100–1000 mcg twice a day (max: 2000 mcg/day)
	Mometasone	1–7 days	1–2 wks		Powder (110, 220 mcg/inhalation) 220–440 mcg one or two times/day (max: 880 mcg/day)

Therapeutic Options: Bronchodilators

- Bronchodilator medications are central to the symptomatic management of COPD.
- Bronchodilators are prescribed on an as-needed or on a regular basis to prevent or reduce symptoms.
- The principal bronchodilator treatments are beta2-agonists, anticholinergics, theophylline or combination therapy.
- The choice of treatment depends on the availability of medications and each patient's individual response in terms of symptom relief and side effects.

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- Long-acting inhaled bronchodilators are convenient and more effective for symptom relief than short-acting bronchodilators.
- Long-acting inhaled bronchodilators reduce exacerbations and related hospitalizations and improve symptoms and health status.
- Combining bronchodilators of different pharmacological classes may improve efficacy and decrease the risk of side effects compared to increasing the dose of a single bronchodilator.

Therapeutic Options: Inhaled Corticosteroids

- Regular treatment with inhaled corticosteroids (*ICS*) improves symptoms, lung function and quality of life and reduces frequency of exacerbations for COPD patients with an $FEV_1 < 60\%$ predicted.
- Inhaled corticosteroid therapy is associated with an increased risk of pneumonia.
- Withdrawal from treatment with inhaled corticosteroids may lead to exacerbations in some patients.

Therapeutic Options: Combination Therapy

- An inhaled corticosteroid combined with a long-acting beta₂-agonist is more effective than the individual components in improving lung function and health status and reducing exacerbations in moderate to very severe COPD.
- Combination therapy is associated with an increased risk of pneumonia.
- Addition of a long-acting beta₂-agonist/inhaled glucocorticosteroid combination to an anticholinergic (tiotropium) appears to provide additional benefits.

Therapeutic Options: Systemic Corticosteroids

- Chronic treatment with systemic corticosteroids should be avoided because of an unfavorable benefit-to-risk ratio.
- Used only during exacerbations for short course

Therapeutic Options: Theophylline

- Theophylline is less effective and less well tolerated than inhaled long-acting bronchodilators and is not recommended if those drugs are available and affordable.
- There is evidence for a modest bronchodilator effect and some symptomatic benefit compared with placebo in stable COPD. Addition of theophylline to salmeterol produces a greater increase in FEV₁ and breathlessness than salmeterol alone.
- Low dose theophylline reduces exacerbations but does not improve post-bronchodilator lung function.

Therapeutic Options: Other Pharmacologic Treatments

- ❖ *Alpha-1 antitrypsin augmentation therapy*: not recommended for patients with COPD that is unrelated to the genetic deficiency.
- ❖ *Mucolytics*: Patients with viscous sputum may benefit from mucolytics; overall benefits are very small.
- ❖ *Antitussives*: Not recommended.
- ❖ *Vasodilators*: Nitric oxide is contraindicated in stable COPD. The use of endothelium-modulating agents for the treatment of pulmonary hypertension associated with COPD is not recommended.

Manage Stable COPD

- Long-acting formulations of beta₂-agonists and anticholinergics are preferred over short-acting formulations.
- Based on efficacy and side effects, inhaled bronchodilators are preferred over oral bronchodilators.
- Long-term treatment with inhaled corticosteroids added to long-acting bronchodilators is recommended for patients with high risk of exacerbations.

Manage Stable COPD: Key Points

- Long-term monotherapy with oral or inhaled corticosteroids is not recommended in COPD.
- The phosphodiesterase-4 inhibitor roflumilast may be useful to reduce exacerbations for patients with $FEV_1 < 50\%$ of predicted, chronic bronchitis, and frequent exacerbations.

Manage Stable COPD: Goals of Therapy

- Relieve symptoms
- Improve exercise tolerance
- Improve health status
- Prevent disease progression
- Prevent and treat exacerbations
- Reduce mortality

} Reduce symptoms

} Reduce risk

Manage Exacerbations

An exacerbation of COPD is:

“an acute event characterized by a worsening of the patient’s respiratory symptoms that is beyond normal day-to-day variations and leads to a change in medication.”

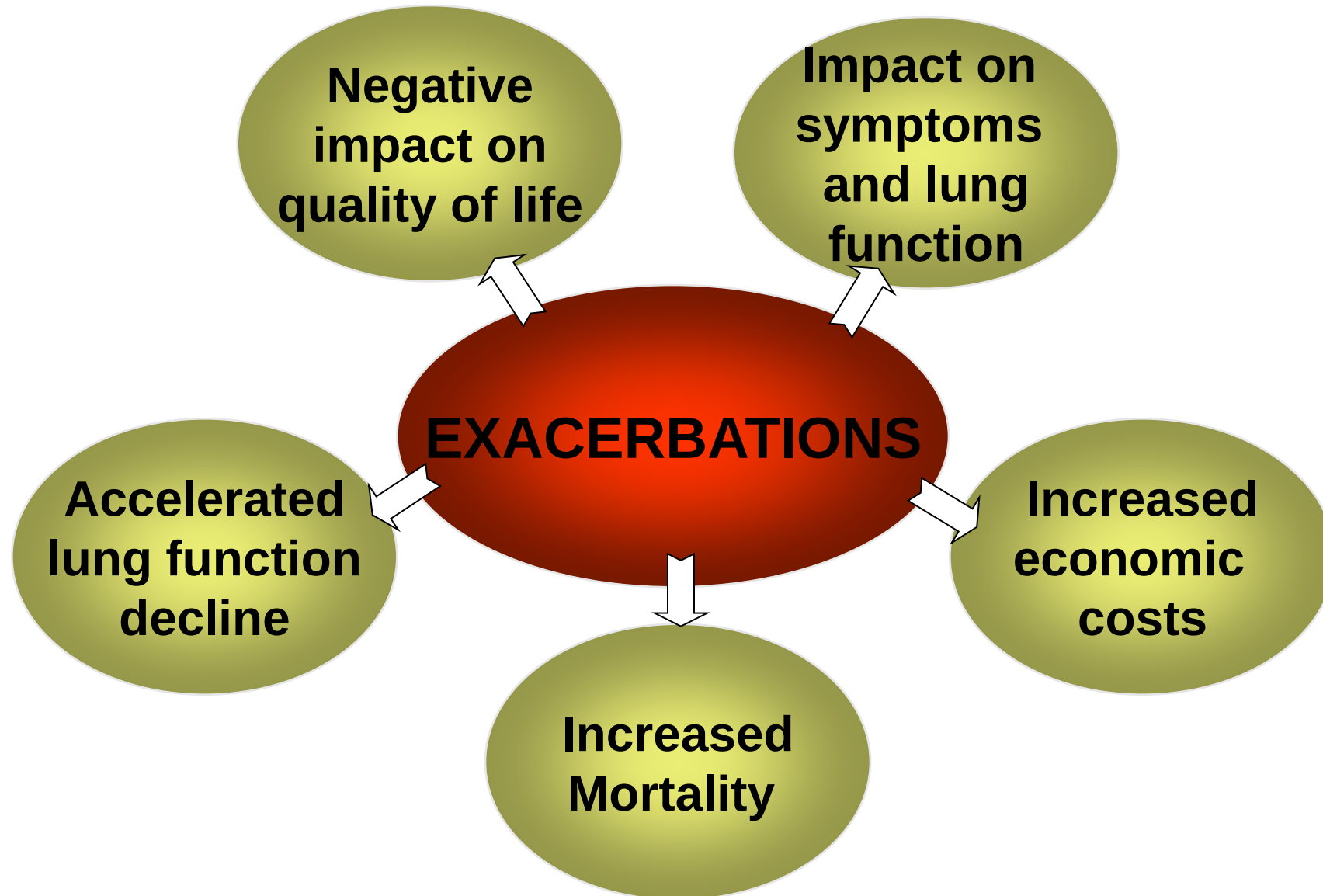
Manage Exacerbations: Key Points

- The most common causes of COPD exacerbations are viral upper respiratory tract infections and infection of the tracheobronchial tree.
- Diagnosis relies exclusively on the clinical presentation of the patient complaining of an acute change of symptoms that is beyond normal day-to-day variation.
- The goal of treatment is to minimize the impact of the current exacerbation and to prevent the development of subsequent exacerbations.

Manage Exacerbations: Key Points

- **SABA** with or without **short-acting anticholinergics** are usually the preferred bronchodilators for treatment of an exacerbation.
- **Systemic corticosteroids** and **antibiotics** can shorten recovery time, improve lung function (FEV_1) and arterial hypoxemia (PaO_2), and reduce the risk of early relapse, treatment failure, and length of hospital stay.
- COPD exacerbations can often be prevented

Consequences of COPD Exacerbations



Manage Exacerbations: Treatment Options

- ***Oxygen:*** titrate to improve the patient's hypoxemia with a target saturation of 88-92%.
- ***Bronchodilators:*** Short-acting inhaled beta₂-agonists with or without short-acting anticholinergics are preferred.
- ***Systemic Corticosteroids:*** Shorten recovery time, improve lung function (FEV₁) and arterial hypoxemia (PaO₂), and reduce the risk of early relapse, treatment failure, and length of hospital stay. A dose of 30-40 mg prednisolone per day for 10-14 days is recommended.

Manage Exacerbations: Treatment Options

❑ Antibiotics should be given to patients with:

❖ Three cardinal symptoms:

- ✓ increased dyspnea,
- ✓ increased sputum volume, and
- ✓ increased sputum purulence.

❖ Who require mechanical ventilation.

Manage Exacerbations: Indications for Hospital Admission

- Marked increase in intensity of symptoms
- Severe underlying COPD
- Onset of new physical signs
- Failure of an exacerbation to respond to initial medical management
- Presence of serious comorbidities
- Frequent exacerbations
- Older age
- Insufficient home support

THANKS